INSIDE STUDIO SP · VST3 · AU

MIX SERIES

MIX AND MASTER IN A 19" RACK

Step-by-step manual

10 plugins · Windows and macOS · VST3 and AU



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What is the MIX SERIES

The mix and master tools, in a 19" rack. Multiband, compressor, limiter with LUFS, console color, transient shaper, de-esser, M/S stereo, reverb, delivery meter and clipper. Each one shows on screen what it's doing.

PLUGIN	WHAT IT'S FOR
PRISMA	Multiband: 4 bands with compressor and saturation.
TORNO	Compressor that shows what it's squeezing.
BARRAGEM	Master limiter with true peak, LUFS and delivery targets.
CONSOLE	Analog channel color: Cristal, Ferro and Brasa.
SOCO	Transient shaper with before and after on screen.
LIXA	De-esser that shows on the spectrum what it removes.
HORIZONTE	Zero-latency M/S stereo imager, with mono bass.
SALA	Clean reverb to put everything in the same space.
MONITOR	LUFS, LRA, true peak, spectrum and correlation.
NAVALHA	4× clipper that shows live what it cuts.

Before you start

ITEM	DETAILS
Formats	VST3 (Windows and Mac) and AU (Mac).
Systems	Windows 10 or 11, 64-bit · macOS 11 Big Sur or newer (Apple Silicon or Intel).
DAWs	Any DAW that loads VST3 or AU: Ableton Live, FL Studio, Logic, Cubase, Bitwig, Reaper, Studio One…

Install and activate

Installation

1. Download the installer from the link in your purchase e-mail (Windows or Mac).
2. **Windows:** run the .exe installer and follow the screens. The plugins go to `C:\Program Files\Common Files\VST3`.
   Mac: open the .dmg, double-click the .pkg installer and follow the screens (VST3 and AU in `/Library/Audio/Plug-Ins`).
3. Close and reopen your DAW. If the plugin doesn't show up, have the DAW rescan its plugins (in Logic: Logic Pro › Settings › Plug-in Manager › "Reset & Rescan Selection").

> **SECURITY WARNING.** The installers are new, so Windows or macOS may show a warning. Windows: "More info" › "Run anyway". Mac: System Settings › Privacy & Security › "Open Anyway". Always download from the link in the e-mail or from the website.

Activation

1. Open any plugin from the pack in your DAW. The activation screen appears.
2. Enter your **purchase e-mail** and your **token** (`ISP-XXXX-…`) and click **ATIVAR** (activate). Internet is only needed at this moment.
3. **Activating one plugin unlocks all 10 plugins in the pack** on that computer.

> **LICENSE.** Valid on up to **2 computers**. Once activated, it works offline for up to **45 days** and renews by itself when the computer goes online. To switch computers: preset menu › `Licença…` (license) › **Desativar este computador** (deactivate this computer), then activate on the new one.

> **WITHOUT ACTIVATION**, the plugin opens but passes the sound straight through, unprocessed.

What applies to all 10 plugins

Working the controls

- **Drag** a knob up and down; the **mouse wheel** also adjusts it.
- **Double-click** resets the control to its default value.
- Each block has a **Macro** (the big knob): on its own it takes the effect from subtle to extreme. The small knobs next to it handle the fine-tuning.
- Knob colors match the drawing on screen: whatever you turn shows up in the same color on the graph.
- The round button at the top is the **bypass** (lit = effect on).

Presets

- Each plugin ships with **20 factory presets**. Click the name to open the list; ◀ ▶ step through them.
- **+** asks for a name and saves the current state. In the menu: *Salvar como novo* (save as new) and, with one of your own presets loaded, *Salvar alterações* (save changes), *Renomear* (rename) and *Apagar* (delete; goes to the trash).
- Performance and listen buttons (such as TAPE STOP, CONGELAR, DELTA, SÓ O CORTE) don't change when you switch presets.

SYSTEM	USER PRESET FOLDER
Mac	`~/Library/Audio/Presets/InsideStudioSP/<PLUGIN>/`
Windows	`%APPDATA%\InsideStudioSP\<PLUGIN>\Presets\`

Meters and latency

Every plugin shows on screen what it's doing (what it cuts, compresses or adds) and has input and output meters. When there is latency (oversampling, lookahead), it shows in the header and is reported to the DAW, which compensates automatically.

> **HOW TO READ THE NEXT PAGES.** Each plugin has its screen, what it does, the control table (name as shown on the plugin, range and what it does) and a usage tip. Display-only areas (graphs and meters) are described in the opening text. Labels shown in Portuguese on the plugin keep their name here, with the English meaning in parentheses the first time.

PRISMA MULTIBAND DYNAMICS + SATURATION

Splits the sound into **4 bands** (4th-order Linkwitz-Riley crossovers, flat sum), each with a compressor and a DRIVE (tanh saturation with level compensation). At the top, the **live output spectrum** with the 4 bands in color: the lines between them are the crossovers and **can be dragged with the mouse**; each band's gain reduction hangs from the top of the band, and the drive, in coral, sits at its foot.



BLOCK	CONTROL	RANGE	WHAT IT DOES
SPECTRUM	**display + crossovers**	20 Hz – 20 kHz	Live output (2048 FFT, 3 dB/octave tilt around 1 kHz to read the highs). Drag a line to move the crossover; double-click resets to default (120 Hz, 1 kHz, 6 kHz). Crossovers always stay in order, with 20% spacing between them
	GR	0 to 24 dB	Band-colored strip hanging from the top: how much the compressor is taking off right now
	drive	—	Coral bar at the foot of the band: effective drive (including the MACRO)
01–04 BANDA (band)	**range and GAIN REDUCTION**	—	Band frequencies and reduction meter (0 to −24 dB)
	SOLO	on/off	Listen only to the soloed bands. Presets don't touch it
	MUTE	on/off	Removes the band from the output. Presets don't touch it
	Thresh	−60 to 0 dB	Level above which the band is compressed (6 dB soft knee)
	Ratio	1:1 to 20:1	How much it compresses above the threshold
	Attack	0.1 to 100 ms	How fast it starts compressing
	Release	10 to 1000 ms	How fast it lets go
	Drive	0 to +24 dB	Soft saturation after the compressor; the level at −12 dBFS stays the same. From 0 to +6 dB it fades in gradually
	Gain	−18 to +18 dB	Band level
05 GLOBAL	**Macro**	0–100%	On all bands at once: threshold down to −12 dB, ratio up to +3 and drive up to +12 dB
	Dry/Wet	0–100%	Dry ↔ processed (parallel; the dry goes through the same crossovers, so there's no phase cancellation)
	Gain	−24 to +12 dB	Output level
	IN / OUT	peak dBFS	Input and output meters

Latency: zero (IIR crossovers, no oversampling); shown in the header and reported to the host.

> **TIP.** Drag the spectrum lines to place the bands and use SOLO to hear one band at a time while you tweak. The GLOBAL Macro tightens all 4 bands together.

TORNO COMPRESSOR

Mix compressor that **shows what it's squeezing**: at the top, the history of the last ~4 s, with the input level in cream rising from the bottom and the gain reduction in coral coming down from the top, plus the THRESHOLD line. The input × output curve shows the knee, the makeup and the current point.



BLOCK	CONTROL	RANGE	WHAT IT DOES
01 COMPRESSÃO (compression)	**Macro APERTO** (squeeze)	0–100%	Lowers the THRESHOLD (down to −24 dB) and multiplies the RATIO (up to 2.5×, max 20:1) together
	Threshold	−60 to 0 dB	Level above which it compresses
	Ratio	1:1 to 20:1	How much it compresses above the threshold
	Knee	0–24 dB	0 = hard knee; more = gradual onset of compression
	SC HPF	Desligado (off), 20–500 Hz	High-pass (12 dB/oct) on the detector only: the low end stops driving the compression
02 ENVELOPE E GAIN	**Attack**	0.1–100 ms	How fast it starts reducing
	Release	10–1000 ms	How fast it lets go
	AUTO RELEASE	on/off	Program-dependent release: an isolated peak releases fast (~40 ms), constant reduction releases slowly (~600 ms)
	Makeup	−12 to +24 dB	Gain after compression
	AUTO	on/off	Adds an estimate to MAKEUP (half the reduction of a 0 dBFS peak); the knob becomes a fine trim
03 OUTPUT	**SC LISTEN**	on/off	Listen to the detector signal (with the SC HPF applied) to find the cutoff. Presets don't touch it
	Dry/Wet	0–100%	Dry ↔ compressed (parallel)
	Gain	−36 to +12 dB	Output level
	IN / OUT	−48 to 0 dBFS	Peak meters

Latency: zero (no lookahead); shown in the header and reported to the host.

> **TIP.** Turn on SC LISTEN and raise the SC HPF until the low end stops driving the compression; then turn SC LISTEN off. For a quick start, just turn APERTO.

BARRAGEM

MASTER LIMITER (TRUE PEAK + LUFS)

Master limiter with **true peak** detection (4× oversampling on the detection path only) and a **loudness** meter (BS.1770 / EBU R128). At the top, the waveform of the last ~4 s: the output in cream, what the dam held back in coral, gain reduction in blue hanging from the top, and the CEILING line. On the left, integrated LUFS with the distance to the chosen target (Spotify, Apple Music, YouTube, Club…).



BLOCK	CONTROL	RANGE	WHAT IT DOES
	Target	12 targets	Click the TARGET box: Spotify −14, Spotify Alto (loud) −11, Apple Music −16, YouTube −14, Tidal −14, Amazon Music −14, Deezer −15, Podcast −16, Club −9, Pista Pesada (heavy dancefloor) −7, TV EBU R128 −23, TV ATSC A/85 −24
01 LIMITE (limit)	**Macro ENCHER** (fill)	0 to +24 dB	Gain before the limiter: pushes the sound into the CEILING
	Ceiling	−12 to 0 dBTP	Maximum output level, also between samples (true peak)
	Release	1 to 1000 ms	How long the gain takes to recover after holding
	AUTO RELEASE	on/off	An isolated peak releases fast (down to 0.3× RELEASE); when the limiter works all the time, it releases slowly (up to 4×) and doesn't pump
	Lookahead	0.5 to 5 ms	Length of the ramp that lowers the gain before the peak: short = more punch, long = cleaner
	Link L/R	0–100%	100% = both channels get the same reduction (the stereo image doesn't move)
03 OUTPUT	**DELTA**	on/off	Listen only to what BARRAGEM is removing. Presets don't touch it
	RESET INTEGRATED	one-shot	Restarts the integrated measurement (and the measured time). Presets don't touch it
	Peaks and meters	—	Plugin INPUT PEAK and OUTPUT PEAK in dBFS (3 s hold) and IN/OUT meters

Latency: fixed, reported to the host = maximum lookahead window (5 ms) + oversampling filter latency (shown in the header). Turning LOOKAHEAD changes only the ramp, not the latency — the host doesn't need to re-compensate. The audio doesn't go through the oversampling filter: what comes out is the sound itself, with gain only, plus a final safety clamp at the CEILING.

> **TIP.** Pick the platform TARGET, play the whole track and raise ENCHER until INTEGRATED reaches the target. Turn on DELTA to hear what's being held back.

CONSOLE

CHANNEL COLOR / CONSOLE SUMMING

Analog console channel color with 2× or 4× oversampling (linear-phase FIR) on the saturation. Three console models, each with its own curve, its own balance of even/odd harmonics, a fixed EQ color and its own noise floor. At the top, the signal going into the channel and the **color it adds**, live; the transfer curve shows the current peak point, and the bars show how much 2nd, 3rd, 4th and 5th harmonic is being generated.



BLOCK	CONTROL	RANGE	WHAT IT DOES
01 CANAL (channel)	**Macro DRIVE**	0–100%	Gain before the curve: 0 to +24 dB. Pushes the channel
	Modelo (model)	Cristal / Ferro / Brasa (crystal / iron / ember)	Channel curve, fixed EQ and noise
	Tone	−6 to +6 dB	Tilt around 700 Hz: + = highs up and lows down
	Crosstalk	0–100%	L↔R bleed between 150 Hz and 7 kHz, up to −16.5 dB
	Noise	Desligado, −96 to −60 dBFS	Model noise floor (added after the compensation)
03 OUTPUT	**COMPARAR** (compare)	on/off	Listen to the dry **at the same volume** as the processed signal (honest A/B). Presets don't touch it
	AUTO	on/off	Auto makeup: the channel comes out at the same level (~0.5 s RMS) it went in; ±24 dB limit
	OS 4×	on/off	4× oversampling (off = 2×, uses less CPU). Presets don't touch it
	Dry/Wet	0–100%	Dry ↔ channel (parallel; the dry gets the same latency)
	Gain	−36 to +12 dB	Output level

Latency: that of the 4× oversampling (shown in the header), reported to the host. At 2× the engine pads the delay up to the 4× value, so changing quality doesn't change latency.

> **TIP.** Put CONSOLE on several channels with the same Modelo so the mix gets one shared color. Turn on COMPARAR to hear the dry at the same volume: that way the comparison is honest.

SOCO TRANSIENT SHAPER

Shapes the **attack** and **sustain** of each hit regardless of level: the detector compares a fast envelope with a slow one (the difference is the hit) and two envelopes with different releases (the difference is the tail). Works on kick, snare, acoustic guitar, bass, loops… At the top, the waveform of the last ~3 s shows BEFORE (ghost) and AFTER overlaid, with what SOCO added or removed painted in that part's color: coral = attack, blue = sustain.



BLOCK	CONTROL	RANGE	WHAT IT DOES
01 SOCO	**Macro ATTACK**	−100% to +100%	More = the start of the hit jumps out (up to +15 dB); less = the hit softens (down to −15 dB)
	Sustain	−100% to +100%	More = the tail grows (room, body; up to +12 dB); less = the tail disappears (dry)
	Foco (focus)	2–50 ms	How long an attack counts as the hit: short = just the click; long = the body of the hit
	Detector	Full · Low · High	Detector-only filter (the sound isn't filtered): Low = low-pass, High = high-pass, 12 dB/octave
	Cutoff	40 Hz–8 kHz	Detector filter frequency (does nothing in Full)
03 OUTPUT	**DELTA**	on/off	Listen only to what SOCO added or removed (with DRY/WET). Presets don't touch it
	Clipper de proteção (safety clipper)	on/off	Soft clipper at the end: clean up to −2.3 dBFS, tops out at −0.3 dBFS
	Dry/Wet	0–100%	Dry ↔ processed (parallel)
	Gain	−36 to +12 dB	Output level (before the clipper)

Latency: zero (the detector doesn't look ahead). Gain is the same on both channels, so the stereo image doesn't move. The IN/OUT meters are peak, in dBFS.

> **TIP.** Positive ATTACK on kick and snare adds punch; negative Sustain dries up the drum room. Turn on DELTA to hear only what SOCO is changing.

LIXA DE-ESSER

Adjustable-band de-esser that **shows what it's removing**: at the top, the live spectrum of the input (gray) and output (cream), with what LIXA removed in coral, and the detection band in amber, with handles on the edges to drag. The analysis panel shows how much it's reducing right now, the band level against the THRESHOLD and the history of the last ~4 s.



BLOCK	CONTROL	RANGE	WHAT IT DOES
01 FAIXA DE DETECÇÃO (detection band)	**Low freq**	1–12 kHz	Lower edge of the band (24 dB/octave high-pass)
	High freq	2–20 kHz	Upper edge of the band (24 dB/octave low-pass); always at least 1.2× above LOW FREQ
	WIDEBAND	on/off	Off: only the band is turned down. On: the whole signal is turned down together (more natural on exposed vocals)
02 GAIN REDUCTION	**Macro RANGE**	0 to 24 dB	The most the S can be turned down
	Threshold	−60 to 0 dB	Band level above which LIXA acts (5:1 ratio, 6 dB knee)
	Link L/R	0–100%	100% = both channels get the same reduction (the stereo image doesn't move)
	Attack	0.1–10 ms	How fast the reduction kicks in
	Release	5–200 ms	How fast the reduction lets go
04 OUTPUT	**SÓ O CORTE** (cut only)	on/off	Listen only to what's being removed (to find the S band). Presets don't touch it
	Gain	−12 to +12 dB	Output level
	Meters	—	PEAK IN and PEAK OUT in dBFS (LEDs and number)

Latency: zero (IIR filters, no look-ahead detector); reported to the host. In band-only mode, with no reduction the output is identical to the input (output = input − band × reduction).

> **TIP.** Turn on SÓ O CORTE and drag the amber band until you hear only the "s" sounds; then turn it off and set RANGE and Threshold. On very exposed vocals, try WIDEBAND.

HORIZONTE STEREO IMAGER

M/S stereo imager with **zero latency**: widens or narrows the stereo, with separate widths for lows and highs, keeps the low end mono below a frequency, and rotates/balances the image. The vectorscope shows the output, with a fading trail; the correlation meter warns when something is going to vanish in mono.



BLOCK	CONTROL	RANGE	WHAT IT DOES
	Correlation	−1 to +1	~300 ms average; below 0 (coral) part of the sound vanishes in mono. MIN holds 3 s. Also shows SIDE/MID (dB) and the measured L/R BALANCE
01 WIDTH	**Macro WIDTH**	0–200%	SIDE (S) gain: 0% = mono, 100% = as it came in, 200% = double
	Low	0–200%	Width below the CROSSOVER (multiplies the macro)
	Crossover	80 Hz–4 kHz	Where lows and highs split (Linkwitz-Riley 24 dB/oct, flat sum)
	High	0–200%	Width above the CROSSOVER (multiplies the macro)
02 BASS E POSIÇÃO (bass and position)	**Mono bass**	Desligado, 20–300 Hz	Zeroes the SIDE (S) below the frequency (Linkwitz-Riley, flat sum)
	Balance	E 100 to D 100 (E = left, D = right)	Lowers the opposite side
	Rotation	−45° to +45°	Rotates the whole image (+ = to the right)
03 OUTPUT	**MONO**	on/off	Mono check: (L+R)/2 on both sides. Presets don't touch it
	SIDE SOLO	on/off	Listen only to S, on both sides. Turning one on turns the other off. Presets don't touch it
	Gain	−24 to +12 dB	Output level
	IN / OUT	dBFS	Input and output peak

Latency: zero (IIR filters only). M goes through the same splitters as S, so both stay in phase. Works only on stereo tracks (stereo in and out).

> **TIP.** Use Mono bass to keep the low end solid in the center and open up only the High. Check with MONO and keep an eye on Correlation: below 0, part of the sound vanishes in mono.

SALA MIX REVERB

Clean, transparent reverb for putting instruments and vocals in the same space (the opposite of MIRAGEM, which is shimmer). An 8-line Hadamard-mixed network, with all-pass diffusers on the input, 16 early reflections per side and a slow, subtle wobble on the lines (removes the metallic ring without turning into chorus). At the top, the **response**: the dry hit, the pre-delay, the reflections and the decay curve, with the current tail line. Below, the **TAIL EQ** with damping and width.



BLOCK	CONTROL	RANGE	WHAT IT DOES
01 ESPAÇO (space)	**Macro SIZE**	0–100%	The 8 lines and the reflections grow together (moving it stretches the room slowly, no clicks)
	Decay	0.2–12 s	RT60: time for the tail to drop 60 dB
	Pre-delay	0–250 ms	Silence before the room responds (used with PRE SYNC on *Livre*)
	Pre sync	Livre (free), 1/64…1/4	Pre-delay as a division of the host tempo (no host: 120 BPM; max 1 s)
	Diffusion	0–100%	Input all-pass: 0 = grainy attack; 100% = smooth tail from the start
	Early ref.	0–100%	Early reflection density: how many of the 16 come in (0 = none)
02 TAIL	**Low cut**	20 Hz–1 kHz	Low cut on the tail (2-pole)
	High cut	1–20 kHz	High cut on the tail (2-pole)
	Damping	0–100%	Filter inside the room: highs die before lows (20 kHz → 1 kHz)
	Width	0–150%	0 = mono tail; 100% = natural; above, wider
	Duck	0–100%	The tail dips (down to −24 dB) while the DRY plays and comes back in the gaps
03 OUTPUT	**SÓ A SALA** (room only)	on/off	Listen 100% wet, to dial in the room. Presets don't touch it
	Dry/Wet	0–100%	Dry ↔ wet
	Gain	−36 to +12 dB	Output level

Latency: zero. The tail reported to the host is 13.5 s (max decay + max pre-delay). Each line's feedback is always below 1 (even with a 12 s decay) and the matrix preserves energy, so the tail always decays; if anything goes NaN, the room resets itself.

> **TIP.** Use one SALA on a return (send) channel for several instruments: they all start sounding in the same room. Duck lowers the tail while the sound plays and lets it bloom in the gaps.

MONITOR LOUDNESS METER

Delivery meter: it **doesn't change the sound** (output equals input, bit for bit, while the checks are off) and has no latency. Measures momentary, short-term and integrated LUFS (BS.1770-4, K-weighting, −70 LUFS and −10 LU gates), LRA (EBU Tech 3342), true peak with 4× oversampling and held maximum, peak and RMS in dBFS, spectrum, phase correlation and goniometer. The presets are **delivery targets** (Spotify, Apple Music, YouTube, EBU R128…): they set the target, the ceiling and the color ranges.



BLOCK	CONTROL	RANGE	WHAT IT DOES
01 TARGET	**Macro TARGET**	−36 to −6 LUFS	Requested integrated loudness (violet line)
	TP ceil	−6 to 0 dBTP	Maximum accepted true peak (coral line on LEVEL)
	Faixa (range)	±0.5 to ±6 LU	Tolerance around the TARGET (violet band)
	LRA max	3 to 30 LU	Maximum accepted LRA
02 CHECAGEM (check)	**MONO**	on/off	Sums L+R on both sides (affects only what you hear)
	MATCH TARGET	on/off	Turns the level down to the TARGET based on the integrated value, like the platforms do; never turns up
	ZERAR (reset)	one-shot	Restarts integrated, LRA, max short-term, max true peak and the time
	CONGELAR (freeze)	on/off	Freezes the display (measurement continues). Not a parameter

Latency: none. The true peak oversampling is analysis only; the audio passes straight through.

> **TIP.** Put MONITOR at the end of the master chain, pick the platform preset (it sets the target, ceiling and ranges), click ZERAR and play the whole track. MATCH TARGET shows how it will sound after the platform's normalization.

NAVALHA CLIPPER

Clipper with 4× oversampling (linear-phase FIR) that **shows what it's cutting**: at the top, the waveform of the last ~4 s, with what passes in cream and what NAVALHA cut in coral, plus the CEILING line. The input × output curve shows the current peak point, and the analysis panel shows how much it's cutting.



BLOCK	CONTROL	RANGE	WHAT IT DOES
01 CORTE (cut)	**Macro DRIVE**	0 to +24 dB	Gain before the clipping: pushes the sound into the CEILING
	Ceiling	−12 to 0 dB	Maximum output level
	Knee	0–100%	0 = straight (hard) clip; more = rounded knee (up to half the ceiling)
	Asymmetry	0–100%	The negative side clips up to 50% earlier: even harmonics, "warm" sound
	Link L/R	0–100%	100% = both channels get the same clipping (the stereo image doesn't move)
03 OUTPUT	**SÓ O CORTE**	on/off	Listen only to what's being removed (to find the sweet spot). Presets don't touch it
	Dry/Wet	0–100%	Dry ↔ clipped (parallel; the dry gets the same latency)
	Gain	−36 to +12 dB	Output level

Latency: that of the oversampling (shown in the header), reported to the host. The output peak may slightly exceed the CEILING between samples, because of the oversampling filter.

> **TIP.** Before the limiter, raise DRIVE until NAVALHA clips only the loudest peaks. SÓ O CORTE lets you hear whether it's taking body as well as peaks.

FAQ and troubleshooting

SITUATION	WHAT TO DO
The plugin doesn't show up in the DAW	Close and reopen the DAW and have it rescan plugins. In Logic: Plug-in Manager › "Reset & Rescan Selection". Make sure you ran the installer for your system.
The sound doesn't change	Check that the pack is activated (preset menu › Licença…). Without activation, the sound passes straight through.
I switched computers	Preset menu › Licença… › Desativar este computador on the old one, then activate on the new one with the same e-mail and token. Valid on up to 2 computers.
I'm offline	Once activated, it works for up to 45 days without internet and renews by itself when the connection is back.
I lost my token	Write to contato@insidestudiosp.com.br with the e-mail used for the purchase.
Did MONITOR change the sound?	No. With MONO and MATCH TARGET off, the output equals the input, bit for bit, with no latency.
Compare with the original	Use the bypass (round button at the top). On CONSOLE, COMPARAR already matches the dry level.

SUPPORT. Questions, suggestions or problems: **contato@insidestudiosp.com.br** · **insidestudiosp.com.br**